

THE STRUCTURE OF MATHEMATICAL REALITY

Phase V — Chapter 17

Open Questions and the Shape of the Unknown

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. You encounter a new mathematical concept in a paper. You have never seen it before. Using the map from this curriculum, describe the procedure you would follow to locate the concept: What field does it belong to? What structure does it preserve? What is it analogous to in fields you know? Where would you look for an accessible introduction?

2. A colleague says: “I have a dataset and I want the simplest possible model.” Advise them, using concepts from at least four different chapters of this curriculum.

3. You discover that two problems you have been working on, one in data analysis and one in experimental design, have the same mathematical structure. What is the correct response? What tools from this curriculum would you use to exploit this discovery?

4. Someone asks you: “Why does math work?” Give your answer. It should take less than two minutes to say out loud. It should be precise, structural, and honest about what remains unknown.

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
